

Dataset Details

A. Domain Context

1. Peer-to-peer used-goods marketplaces
2. One-shot, short (~12-turn) text negotiations
3. Primary task: price bargaining of smartphones/electronics.

Primary Biases:-

1. Anchoring - first numbers strongly shape final prices
2. Confirmation - selective evidence-seeking
3. Overconfidence - excessive certainty and rigidity

B. Five Decision Factors

1 --- Customer Vulnerability (Language-Based) : Detected through budget stress, urgency, negative sentiment, and anxiety markers.

Scoring (0–1):

1. Financial pressure +0.30 (CVS "Purchasing Power" dimension)
2. Urgency/time pressure +0.30 (LIWC-22 Time category)
3. Strong negative affect +0.20 (LIWC-22 Negative Emotion >3%)
4. Uncertainty/anxiety language +0.20 (LIWC-22 Anxiety category >1.2%)

Threshold: $\geq 0.6 \rightarrow$ Mandatory mitigation (≥ 1 FCA vulnerability driver)

2 --- Customer Sophistication (Language + Behavior)

Indicator	Signals
Question Quality	Specific comparative or technical questions show sophistication
Language Complexity	Diverse vocabulary and conditional phrasing indicate expertise
Market Awareness	References to comparable prices and alternatives show research
First Offer Strategy	Lower first offers relative to asking price show anchor control

Scoring (0–1):

1. First offer: +0.35 (Galinsky et al. (2009): explains 50-85% of outcomes)
2. Question quality: +0.30 (Harvard Negotiation Project)
3. Language complexity: +0.20 (LIWC-22 cognitive markers)
4. Market awareness: +0.15 (BATNA/comparable mentions)

Threshold: $\geq 0.6 \rightarrow$ Sophisticated enough for ethical leverage

3 --- Bias Strength (Numerical Anchoring Index)

Anchoring Strength = $1 - (\text{Actual Adjustment} / \text{Rational Adjustment})$

Rule: Strong anchoring that harms the customer $\rightarrow$ mitigation

Index Range	Classification	Decision	Empirical Basis
0.70 – 1.00	Strong anchoring	Mitigate if harms customer	Adjustment <30% of rational (Epley & Gilovich)
0.50 – 0.70	Moderate anchoring	Context-dependent	Typical population range
0.00 – 0.50	Weak anchoring	Neutral	Relatively rational behavior

Asking Price (P) = Price listed by seller
Anchor (A) = First price mentioned by customer
Fair Value (F) = Estimated market value (typically 75% P)
Current Offer (O) = Customer's latest offer
Rational Adjustment = $|F - A|$
Actual Adjustment = $|O - A|$

4 --- Harm Direction (Numerical Fairness Check)

Customer Offer vs Fair Value	Effect	Decision
> Fair Value	overpayment	Mitigate
$\leq$ Fair Value	benefit	Customer advantaged

Tolerance ($\pm 10\%$) Justification: Baye et al. (2006) found online retail price dispersion SD = 10-15%; used goods = 15-25%. The $\pm 10\%$ threshold captures normal market friction.

5 ---Vocal Stress & Emotional Instability Index (VSEI):
Minimum 16kHz sampling rate for prosodic analysis

Scoring system (0-1):

****VSEI Score Validation**:**

- Calculate VSEI from prosodic features (as per your paper)
- Compare against human-rated stress levels
- ****Correlation metric**:** Pearson correlation between VSEI and human ratings

D. Reward Design Principles

- 1. Balance deal value, customer fairness, interaction quality
- 2. Penalize outcomes where customer is harmed
- 3. Weight losses to customer penalised more heavily than gains to agent
- 4. Deduct for actions that compromise autonomy/fairness

E. Dataset Specification (5,000 Dialogues)

Attribute	Value
Dialogues	5,000
Turns per dialogue	~12
Domain	Used smartphones
Purpose	Train bias-aware negotiation agents

Scenario Distribution balanced across:-

- 1. Vulnerability (low/high)
- 2. Sophistication (low/high)
- 3. Anchor direction (harms/benefits customer)

8 balanced scenarios × 625 each

Persona Requirements:-

- 1. Buyers: college student, urgent replacement seeker, first-time buyer, enthusiast, bargain hunter, etc.
- 2. Sellers: casual seller, upgrader, urgent mover, reseller

Dialogue Requirements:-

- 1. ≥3 customer price offers
- 2. ≥2 questions
- 3. Vulnerability signals where applicable
- 4. Market comparison signals for sophisticated personas
- 5. Customer has the bias/vulnerability/sophistication/harm direction/VSEI not vice versa.

C. Decision Logic

1. Mandatory Mitigation:-
High vulnerability, Low sophistication, Strong anchoring that harms customer

2. Ethical Leverage Allowed:-
Low vulnerability, High sophistication, Customer’s position is neutral or benefits them

Dataset Quality Evaluation:

Automatic Evaluation Metrics:---

Lexical Diversity and Variability:

variability -> BLEU-1, METEOR. (DEAL)

lexical diversity and low redundancy -> BLEU-1/2, METEOR, ROUGE-L, DISTINCT-1/2, Self-BLEU-1/2. (PACT)

Semantic diversity across various meanings and contexts -> Sem-Ent metric (PACT)

To assess 18 and 4 distinct aspects of dialogue quality respectively, ensuring the synthetic interactions mimicked human-like naturalness -> FED (Fine-Grained Evaluation of Dialog) and G-Eval metrics (PACT)

Manual Evaluation and Human Assessment:---

Human experts reviewed the datasets to ensure they met domain-specific requirements and qualitative standards:

- Quality and Efficacy Scales:

- NEGOCHAT utterances were rated on a 1-3 scale for negotiation efficacy.

- DEAL was assessed on a 1-3 scale for Naturalness (Nat), Engagingness (Eng), Fairness (Fair), and Faithfulness (Faith)

- PACT utilized a 1-4 scale to evaluate Personality Relevance (PR), Bargaining Efficacy (BE), Outcome Fairness (OF), Engagingness (E), Coherence (C), Humanness (H), Fluency (F).

- PSYCON evaluated for humanness, user profile consistency, and psychotherapeutic approach consistency using a 1-3 scale.

- Turing Test: Turing test - human evaluators attempted to distinguish between AI-generated and human-written dialogues. (DEAL)

- Filtering and Cleaning: Experts reviewed DEAL for fluency (1-5 scale) and naturalness (0-2 scale) to filter out disfluent or inauthentic utterances.

Inter-Rater Reliability Metrics

- Kappa Statistics: Kappa agreement ratios or Fleiss Kappa (κ) to measure the reliability of human judgments across all datasets. NEGOCHAT - inter-subject Kappa of 81.6% for quality assessment.

- Agreement Ratios: General agreement ratios to validate consistency of post-editing + filtering processes, 84.9% ratio observed for fluency in the DEAL dataset

DEAL - Trip Negotiator
PACT - We argue to agree
PSYCON - E-Therapist
GENTEEL (IND, NEGOCCHAT) - Genteel Negotiator

WHAT WE CAN DO---

For both text, speech:

1. Kappa/agreement ratios

Kappa Statistics for manual audio quality ratings
Agreement Ratios for emotion/naturalness judgments

2. Manual Evaluation using 1-3/1-4/1-5 scale. (maybe PACT type parameters for both text+speech)

Naturalness (1-5 scale):
Does the speech sound human-like?, Are prosodic patterns natural?

Audio Clarity (1-5 scale):
Is the audio clear without artifacts? Background noise acceptability

Emotional Authenticity (1-5 scale):
Do expressed emotions sound genuine?, Appropriate emotional intensity

Prosodic Appropriateness (1-5 scale):
Stress patterns match semantic content, Intonation matches dialogue context (questions vs. statements)

Turing Test for Speech:
Can evaluators distinguish synthetic vs. real human speech?

Contextual Emotion Match (1-5):
Does vocal emotion match dialogue content? Appropriate stress/anger/urgency in negotiation context

For only text: (automatic evaluation)

1. Any 1/2 of BLEU, MOTOR, FED, G-Eval, Sem-Ent

For only speech:

1. Emotion Recognition Model Validation:

- Use pre-trained SER models to classify emotions
- Compare predicted emotions vs. ground truth labels
- Metrics: Accuracy, F1-score, Confusion Matrix
- Validate if stress/anger/anxiety are detectable

Tools:

Praat - comprehensive prosodic analysis

OpenSMILE - automatic audio feature extraction

librosa (Python) - audio processing and feature extraction

parselmouth (Python) - Praat functionality in Python

Pre-trained SER (Speech Emotion Recognition) models for emotion validation